

IIT Foundation Module for Class VIII

Physics

- Theme-1: Measurements and Experimentation
 1. International System of Units, the required SI units with correct symbols are given at the end of this syllabus. Other commonly used system of units – fps and cgs.
 2. Measurements using common instruments, Vernier callipers and micro-metre screw gauge for length, and simple pendulum for time.
- Theme-2: Motion in One Dimension
 1. Scalar and vector quantities, distance, speed, velocity, acceleration; graphs of distance-time and speed-time; equations of uniformly accelerated motion with derivations.
- Theme-3: Laws of Motion
 1. Contact and non-contact forces; cgs & SI units.
 2. Newton's First Law of Motion (qualitative discussion) introduction of the idea of inertia, mass and force.
 3. Newton's Second Law of Motion (including $F=ma$); weight and mass.
 4. Newton's Third Law of Motion (qualitative discussion only); simple examples.
 5. Gravitation
- Theme-4: Fluids
 1. Change of pressure with depth (including the formula $p=h\rho g$); Transmission of pressure in liquids; atmospheric pressure.
 2. Buoyancy, Archimedes' Principle; floatation; relationship with density; relative density; determination of relative density of a solid.
- Theme-5: Heat and Energy
 1. Concepts of heat and temperature
 2. Anomalous expansion of water; graphs showing variation of volume and density of water with temperature in the 0 to 10°C range. Hope's experiment and consequences of anomalous expansion.
 3. Energy flow and its importance
 4. Energy sources

5. Global warming and Green House effect

○ Theme-6: Light

1. Reflection of light; images formed by a pair of parallel and perpendicular plane mirrors
2. Spherical mirrors; characteristics of image formed by these mirrors. Uses of concave and convex mirrors. (Only simple direct ray diagrams are required).

○ Theme-7: Sound

1. Nature of Sound waves. Requirement of a medium for sound waves to travel; propagation and speed in different media; comparison with speed of light.
2. Infrasonic, sonic, ultrasonic frequencies and their applications.

Theme-8: Electricity and Magnetism

1. Simple electric circuit using an electric cell and a bulb to introduce the idea of current (including its relationship to charge); potential difference; insulators and conductors; closed and open circuits; direction of current (electron flow and conventional)
2. Induced magnetism, Magnetic field of earth. Neutral points in magnetic fields.
3. Introduction of electromagnet and its uses.